INDIAN ROADS CONGRESS & MoRTH SPECIAL TECHNICAL SPECIFICATIONS
SPECIAL GUIDELINES FOR DEEP FOUNDATIONS IN RIVERBED ALLUVIAL STRATA

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Applicable standard: IRC:78-2014 & MoRTH 5th Revision Section 2200

1. SCOPE:
These specifications govern the design, execution, and quality control of Bored Cast-in-Situ Reinforced Cement Concrete (RCC) Piles and Well Foundations in perennial river basins with high scour potential, including the Brahmaputra and Barak river basins.

2. FOUNDATION DEPTH & SCOUR CALCULATION:
The minimum depth of foundation below maximum scour depth shall be calculated in accordance with IRC:5 and Lacey's Regime Equation:
- Mean Scour Depth: R = 0.473 * (Q / f)^(1/3)
- Maximum Scour Depth under Pier: D_pier = 2.0 * R
- Minimum Pile Embedment: The pile termination level must penetrate at least 1/3 of the maximum scour depth into firm bearing strata with Standard Penetration Test (SPT) N-value exceeding 35.

3. CONCRETE AND REINFORCEMENT REQUIREMENTS:
- Minimum Grade of Concrete for Submerged Piles: M35 with minimum cement content of 400 kg/m3 and slump of 150mm to 180mm using plasticizers.
- Clear Cover to Reinforcement: 75mm minimum for deep piles cast in underwater saline/muddy conditions.
- Minimum Longitudinal Steel: 0.4% to 0.8% of gross cross-sectional area of pile shaft.

4. BEARING PEDESTALS & SEISMIC ARRESTORS:
All bridge piers must be provided with Pot-PTFE or Elastomeric bearings conforming to IRC:83. Transverse seismic arrestor blocks of minimum 400mm thickness must be provided on pier caps to prevent superstructure dislodgement during Richter magnitude 7.0+ earthquake events.